

1. For a nucleus ${}_Z^AX$ having mass number A and atomic number Z
 - A. The surface energy per nucleon $(b_s) = -a_1 A^{\frac{2}{3}}$.
 - B. The Coulomb contribution to the binding energy $b_c = -a_2 \frac{Z(Z-1)}{A^{\frac{1}{3}}}$.
 - C. The volume energy $b_v = a_3 A$
 - D. Decrease in the binding energy is proportional to surface area.
 - E. While estimating the surface energy, it is assumed that each nucleon interacts with 12 nucleons. (a_1 , a_2 and a_3 are constants)

Choose the most appropriate answer from the options given below:

[2023 (08 Apr Shift 1)]

 - (1) B, C, E only
 - (2) C, D only
 - (3) A, B, C, D only
 - (4) B, C only
2. A nucleus with mass number 242 and binding energy per nucleon as 7.6 MeV breaks into two fragment each with mass number 121. If each fragment nucleus has binding energy per nucleon as 8.1 MeV, the total gain in binding energy is _____ MeV.
[2023 (08 Apr Shift 1)]
3. A radio active material is reduced to $\frac{1}{8}$ of its original amount in 3 days. If 8×10^{-3} kg of the material is left after 5 days the initial amount of the material is _____
[2023 (08 Apr Shift 2)]

 - (1) 40 g
 - (2) 32 g
 - (3) 64 g
 - (4) 256 g

4. The decay constant for a radioactive nuclide is $1.5 \times 10^{-5} \text{ s}^{-1}$. Atomic weight of the substance is 60 g mole $^{-1}$, $N_A = 6 \times 10^{23}$. The activity of 1.0 μg of the substance is _____ $\times 10^{10}$ Bq.
[2023 (10 Apr Shift 1)]
5. The half life of a radioactive substance is T . The time taken, for disintegrating $\left(\frac{7}{8}\right)^{\text{th}}$ part of its original mass will be: _____
[2023 (10 Apr Shift 2)]

 - (1) $2T$
 - (2) $3T$
 - (3) T
 - (4) $8T$

6. Two radioactive elements A and B initially have same number of atoms. The half life of A is same as the average life of B . If λ_A and λ_B are decay constants of A and B respectively, then choose the correct relation from the given options.
[2023 (11 Apr Shift 1)]

 - (1) $\lambda_A \ln 2 = \lambda_B$
 - (2) $\lambda_A = \lambda_B$
 - (3) $\lambda_A = \lambda_B \ln 2$
 - (4) $\lambda_A = 2\lambda_B$

7. A common example of alpha decay is

$${}_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He} + Q$$

Given:

$${}_{92}^{238}\text{U} = 238.05060 \text{ u}$$

$${}_{90}^{234}\text{Th} = 234.04360 \text{ u}$$

$${}_2^4\text{He} = 4.00260 \text{ u and } 1\text{u} = 931.5 \frac{\text{MeV}}{c^2}$$

The energy released (Q) during the alpha decay of ${}_{92}^{238}\text{U}$ is _____ MeV.
[2023 (12 Apr Shift 1)]
8. ${}_{92}^{238}\text{A} \rightarrow {}_{90}^{234}\text{B} + {}_2^4\text{D} + Q$

In the given nuclear reaction, the approximate amount of energy released will be:

[Given, mass of ${}_{92}^{238}\text{A} = 238.05079 \times 931.5 \text{ MeV } c^{-2}$, mass of ${}_{90}^{234}\text{B} = 234.04363 \times 931.5 \text{ MeV } c^{-2}$, mass of ${}_2^4\text{D} = 4.00260 \times 931.5 \text{ MeV } c^{-2}$]
[2023 (13 Apr Shift 1)]

 - (1) 3.82 MeV
 - (2) 5.9 MeV
 - (3) 2.12 MeV
 - (4) 4.25 MeV

9. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in the range 30 to 170.

Reason R : Nuclear force is short ranged.

In the light of the above statements, choose the **correct** answer from the options given below

[2023 (13 Apr Shift 2)]

- (1) **A** is false but **R** is true
- (2) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (3) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**
- (4) **A** is true but **R** is false.

10. The half-life of a radioactive nucleus is 5 years. The fraction of the original sample that would decay in 15 years is :

[2023 (15 Apr Shift 1)]

- (1) $\frac{1}{8}$
- (2) $\frac{1}{4}$
- (3) $\frac{7}{8}$
- (4) $\frac{3}{4}$

ANSWER KEYS

1. (2) 2. (121) 3. (4) 4. (15) 5. (2) 6. (3) 7. (4) 8. (4)
9. (2) 10. (3)

1. (2)

The formula to calculate the radius of a nucleus is given by

$$R = R_0 A^{\frac{1}{3}} \dots (1)$$

Hence, the volume energy can be written as

$$b_v = C_1 R^3$$

$$= a_3 \left(A^{\frac{1}{3}} \right)^3$$

$$= a_3 A$$

The nuclear binding energy is the minimum energy that is required to disassemble the nucleus of an atom into its constituent protons and neutrons.

Also, the energy of the nucleus is negative with regard to the energy of the particles pulled apart to infinite distance, because energy must be utilised to split a nucleus into its individual protons and neutrons. It is proportional to the surface area.

Hence, statements C and D are most appropriate.

2. (121)

Binding energy is given by $E = \Delta mc^2$

where Δm is the mass defect.

The energy per nucleon of the nucleus having mass number 242 is 7.6 MeV.

The initial binding energy is,

$$BE = 242 \times 7.6 \text{ MeV}$$

The energy per nucleon of the nucleus with mass number 121 is 8.1 MeV.

Therefore, binding energy is

$$BE' = 2(121 \times 8.1 \text{ MeV})$$

The gain in the binding energy is

$$BE' - BE = (8.1 - 7.6) \times 242 \text{ MeV} = 121 \text{ MeV}$$

$$= 121 \text{ MeV}$$

3. (4)

The initial and final amount of the radioactive substance are related by the formula

$$\frac{N}{N_0} = \left(\frac{1}{2} \right)^n \dots (1)$$

where, n is the number of half lives.

Substitute $\frac{1}{8}$ for $\frac{N}{N_0}$ into equation (1) and solve to calculate the value of n .

$$\frac{1}{8} = \left(\frac{1}{2} \right)^n$$

$$\Rightarrow \left(\frac{1}{2} \right)^3 = \left(\frac{1}{2} \right)^n$$

$$\Rightarrow n = 3$$

The half-life of the radioactive element can be calculated as follows:

$$\frac{t}{t_{1/2}} = n \dots (2)$$

Substitute the values of the known parameters into equation (2) and solve to calculate the half-life.

$$\frac{3 \text{ days}}{t_{1/2}} = 3$$

$$\Rightarrow t_{1/2} = 1 \text{ day}$$

The initial and the final mass of the radioactive element are related by

$$\frac{m}{m_0} = \left(\frac{1}{2} \right)^{t/t_{1/2}} \dots (3)$$

Substitute the values of the known parameters into equation (3) and solve to calculate the initial mass of the substance.

$$\frac{8 \times 10^{-3} \text{ kg}}{m_0} = \left(\frac{1}{2} \right)^5$$

$$\Rightarrow m_0 = 2^5 \times 8 \times 10^{-3} \text{ kg}$$

$$= 256 \text{ g}$$

4. (15)

It is given that

$$\lambda = 1.5 \times 10^{-5} \text{ s}^{-1}$$

$$\text{Number of moles is } \frac{1 \times 10^{-6}}{60} = \frac{10^{-7}}{6}$$

Number of atoms = number of moles $\times N_A$

$$N_0 = \frac{10^{-7}}{6} \times 6 \times 10^{23} = 10^{16}$$

Hence, at $t = 0$, $A = A_0 = N_0 \lambda$, the activity can be given by

$$A = N_0 \lambda e^{-\lambda t}$$

$$\Rightarrow A = 10^{16} \times 1.5 \times 10^{-5} \times 1 = 15 \times 10^{10} \text{ Bq}$$

5. (2)

It is given that $t_{\frac{1}{2}} = T$.

After disintegration of $\left(\frac{7}{8}\right)^{\text{th}}$ part, $\left(\frac{1}{8}\right)^{\text{th}}$ part is only left.

Using the decay formula,

$$N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{t_{\frac{1}{2}}}}$$

$$\Rightarrow \frac{N_0}{8} = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T}}$$

$$\Rightarrow -\ln 8 = -\frac{t}{T} \ln 2$$

$$\Rightarrow 3 \ln 2 = \frac{t}{T} \ln 2$$

$$\Rightarrow t = 3T$$

6. (3)

It is given that

$$T_{\frac{1}{2}}(A) = T_{av}(B)$$

The half life of a substance is given by $T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$

So,

$$\frac{\ln 2}{\lambda_A} = \frac{1}{\lambda_B}$$

$$\Rightarrow \lambda_A = \lambda_B \ln 2$$

7. (4)

The formula to calculate the Q-value of the given nuclear reaction can be written as

$$Q = (m_U - m_{Th} - m_{He})c^2 \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the required Q-value for the reaction.

$$Q = (238.05060 \text{ u} - 234.04360 \text{ u} - 4.00260 \text{ u})c^2$$

$$= 0.0044c^2 \times 931.5 \text{ MeV}/c^2$$

$$\approx 4 \text{ MeV}$$

8. (4)

The value of Q is

$$Q = (m_A - m_B - m_D) \times 931.5 \text{ MeV}$$

$$\Rightarrow Q = (238.05079 - 234.04363 - 4.00260) \times 931.5 \text{ MeV}$$

$$\Rightarrow Q = 4.25 \text{ MeV}$$

9. (2)

The binding energy for stable nuclei is always a positive number, as the nucleus must gain energy for the nucleons to move apart from each other.

Nucleons are attracted to each other by the strong nuclear force. Binding energy per nucleon is almost constant in the range of 30 – 170. This is because nuclear energy is a short range force. It operates in the femto metre region.

Hence, this option is the correct one.

10. (3)

The formula to calculate the number of half lives covered by the radioactive material is given by

$$n = \frac{t}{T_{1/2}} \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the number of half lives covered.

$$n = \frac{15 \text{ years}}{5 \text{ years}} = 3$$

The relation between the amount of radioactive substance after certain time can be written as

$$N = N_0 2^{-n} \dots (2)$$

Substitute the value of n into equation (2) to obtain the fraction of the remaining substance.

$$N = N_0 2^{-3} \Rightarrow \frac{N}{N_0} = \frac{1}{8}$$

Hence, the amount of the decayed substance can be written as

$$1 - \frac{N}{N_0} = 1 - \frac{1}{8} = \frac{7}{8}$$